

IMAT MOCK EXAM 2 — "Bioenergetics Spine"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 47/90.0 · Mock exam · answer key + worked solutions at the end

Section A - Logical Reasoning & Reading (Q1-9)

1. "Antibiotic resistance arises when bacteria acquire genetic changes — by mutation or horizontal gene transfer — that reduce susceptibility to a drug. Resistant strains survive treatment and proliferate, so each course of antibiotics applies selective pressure favouring resistance. Resistance spreads faster in populations with high antibiotic use."

Which is the best-supported conclusion?

- A) Reducing unnecessary antibiotic use would slow the spread of resistance.
- B) Antibiotic use creates genetic mutations directly.
- C) All bacteria become resistant after one course.
- D) Resistance is independent of selection pressure.
- E) Resistance cannot arise without horizontal gene transfer.

2. "Homeostasis refers to the maintenance of a constant internal environment despite external changes. Core variables such as body temperature, blood pH, and blood glucose are regulated by negative feedback loops: a sensor detects deviation, a control centre orchestrates a response, and an effector returns the variable to its set point."

Which is consistent with the passage?

- A) Homeostasis requires no sensor.
- B) Negative feedback returns a variable toward its set point.
- C) A positive-feedback loop is the main homeostatic mechanism.
- D) Body temperature is not a homeostatically regulated variable.
- E) The effector's job is to push the variable further from the set point.

3. (General knowledge) The Italian Renaissance anatomist who published *De humani corporis fabrica* (1543), correcting many Galenic errors through direct dissection, was:

- A) Leonardo da Vinci
- B) Marcello Malpighi
- C) Andreas Vesalius
- D) Luigi Galvani
- E) William Harvey

4. Four students — P, Q, R, S — are ranked in an exam. P scored higher than Q but lower than R. S scored lower than Q. Who scored lowest?

- A) P
- B) R
- C) Q
- D) Cannot be determined
- E) S

5. "If it rains, the match is cancelled." The match was not cancelled. Therefore:

- A) The teams refused to play.
- B) It rained.
- C) The match was cancelled.
- D) It did not rain.
- E) It might have rained.

6. A machine applies the rule: take a number, square it, then subtract 4. If the output is 32, the input could be:

- A) 4 B) 5 C) 6 D) 7 E) 8

7. At 9:45, what is the angle between the hour and minute hands of a clock?

- A)** 0° **B)** 15° **C)** 22.5° **D)** 30° **E)** 45°

8. A is the father of B. B is the sister of C. How is A related to C?

- A)** Grandfather **B)** Uncle **C)** Cousin **D)** Brother **E)** Father

9. Complete the series: 1, 4, 9, 16, 25, ?

- A)** 30 **B)** 33 **C)** 36 **D)** 49 **E)** 64

Section B - Biology (Q10-32)

10. Glycolysis occurs in the:

- A)** Nucleus
B) Inner mitochondrial membrane
C) Mitochondrial matrix
D) Cytosol
E) Chloroplast stroma

11. Under anaerobic conditions in human muscle, pyruvate is converted to:

- A)** Lactate (lactic acid)
B) Acetate
C) Acetyl-CoA
D) Glucose
E) Ethanol and CO₂

12. The net reaction of aerobic respiration (per glucose) produces approximately how many ATP in eukaryotes (modern estimate)?

- A)** 2 **B)** 4 **C)** 6-8 **D)** 30-32 **E)** 100

13. Which coenzyme carries electrons from glycolysis and the Krebs cycle to the electron transport chain?

- A)** Coenzyme A
B) NAD⁺ / NADH and FAD / FADH₂
C) ADP
D) ATP
E) AMP

14. The proton-motive force across the inner mitochondrial membrane is generated by:

- A)** Pumping of H⁺ from matrix to intermembrane space by electron transport complexes
B) Pyruvate transport
C) Diffusion of O₂
D) Ca²⁺ release
E) ATP hydrolysis

15. Which metabolic pathway occurs in both cellular respiration and photosynthesis?

- A)** Calvin cycle
B) The Krebs cycle
C) Light-dependent reactions
D) Glycolysis-like reactions of the Calvin cycle only
E) Electron transport chain with chemiosmosis

16. In which organelle does photosynthesis take place in plant cells?

- A)** Peroxisome **B)** Chloroplast **C)** Mitochondrion **D)** Nucleus **E)** Vacuole

- 17.** The light-dependent reactions of photosynthesis produce:
- A)** ATP, NADPH, and O₂
 - B)** Glucose and O₂
 - C)** Pyruvate
 - D)** Lactate
 - E)** CO₂ and H₂O
- 18.** Which blood vessels carry oxygenated blood from the lungs to the heart?
- A)** Coronary arteries
 - B)** Pulmonary veins
 - C)** Vena cavae
 - D)** Pulmonary arteries
 - E)** Aorta
- 19.** Gas exchange in the lungs occurs across the walls of the:
- A)** Bronchi **B)** Trachea **C)** Aorta **D)** Alveoli **E)** Pharynx
- 20.** Which cells of the gastric glands secrete hydrochloric acid?
- A)** Enterocytes
 - B)** Mucous neck cells
 - C)** Goblet cells
 - D)** Chief cells
 - E)** Parietal (oxyntic) cells
- 21.** Bile, which emulsifies fats, is produced by the __ and stored in the __.
- A)** Gallbladder / liver
 - B)** Pancreas / gallbladder
 - C)** Stomach / pancreas
 - D)** Liver / gallbladder
 - E)** Small intestine / liver
- 22.** The basic functional unit of the kidney is the:
- A)** Sarcomere **B)** Alveolus **C)** Nephron **D)** Hepatocyte **E)** Neuron
- 23.** Filtration of blood in the nephron occurs at the:
- A)** Collecting duct
 - B)** Glomerulus (Bowman's capsule)
 - C)** Loop of Henle
 - D)** Renal pelvis
 - E)** Distal convoluted tubule
- 24.** Which tissue lines the inner surface of the stomach and intestines?
- A)** Simple columnar epithelium
 - B)** Simple cuboidal epithelium
 - C)** Stratified squamous epithelium
 - D)** Pseudostratified ciliated epithelium
 - E)** Transitional epithelium
- 25.** Smooth muscle is found in the walls of all of the following EXCEPT:
- A)** The uterus
 - B)** The bronchi
 - C)** Blood vessels
 - D)** The intestines
 - E)** Skeletal attachments to bone (tendons are connective tissue, not muscle)

26. Which connective tissue cell type produces the fibres of regular dense connective tissue (e.g., tendons)?

- A)** Erythrocytes **B)** Fibroblasts **C)** Osteocytes **D)** Chondrocytes **E)** Adipocytes

27. The primary stimulus for ventilation during moderate exercise is the rise in:

- A)** Blood O_2
B) Blood nitrogen
C) Atmospheric pressure
D) Blood CO_2 (and H^+), sensed by central and peripheral chemoreceptors
E) Body temperature only

28. Insulin promotes which of the following in liver and muscle?

- A)** Glycogenolysis and gluconeogenesis
B) Lipolysis
C) Glycogen synthesis (glycogenesis) and glucose uptake
D) Ketogenesis
E) Glucose release into blood

29. The long bone of the thigh is the:

- A)** Tibia **B)** Fibula **C)** Humerus **D)** Patella **E)** Femur

30. A red blood cell placed in a hypertonic solution will:

- A)** Shrink (crenate)
B) Divide
C) Remain unchanged
D) Swell and burst
E) Become a leucocyte

31. The phase of mitosis in which chromosomes align at the cell's equator is:

- A)** Anaphase **B)** Metaphase **C)** Telophase **D)** Cytokinesis **E)** Prophase

32. Which of the following is the correct base-pairing rule in DNA?

- A)** A – U and G – C (this is DNA-RNA, not DNA-DNA)
B) A – C and G – T
C) A – G and T – C
D) A – T and G – C
E) A – A and T – T

Section C - Chemistry (Q33-47)

33. Balance: $__ Al + __ O_2 \rightarrow __ Al_2O_3$. The coefficient of O_2 is:

- A)** 1 **B)** 2 **C)** 3 **D)** 4 **E)** 6

34. What is the oxidation number of Mn in $KMnO_4$?

- A)** –1 **B)** +2 **C)** +4 **D)** +6 **E)** +7

35. 25.0 mL of 0.10 M HCl is neutralised by 0.05 M NaOH. What volume of NaOH is required?

- A)** 5 mL **B)** 12.5 mL **C)** 25 mL **D)** 50 mL **E)** 100 mL

36. A buffer solution resists changes in pH when small amounts of acid or base are added. A typical buffer consists of:

- A)** A weak acid and its conjugate base (or weak base and its conjugate acid)
B) A strong acid and a strong base
C) Pure water
D) Two strong acids
E) A strong acid and its conjugate base

37. Which has the highest electronegativity?
A) F B) Cs C) Na D) Cl E) O
38. The shape of a methane (CH_4) molecule is:
**A) Linear
B) Tetrahedral
C) Octahedral
D) Pyramidal
E) Trigonal planar**
39. At STP, what volume is occupied by 1 mole of an ideal gas? (Use standard value.)
A) 1.0 L B) 6.02 L C) 11.2 L D) 22.4 L E) 24.0 L
40. Calculate the molar mass of CaCO_3 . (Ca = 40, C = 12, O = 16.)
A) 56 g/mol B) 68 g/mol C) 84 g/mol D) 100 g/mol E) 116 g/mol
41. Which has the lowest pH?
**A) 0.01 M HCl
B) 0.1 M NaOH
C) Pure water
D) 0.1 M CH_3COOH (acetic acid, weak)
E) 0.1 M HCl**
42. Which of the following is an aromatic compound?
A) Ethene B) Ethanol C) Benzene D) Acetylene E) Methane
43. Which functional group is present in aldehydes?
**A) $-\text{CHO}$
B) $-\text{C}=\text{O}$ (internal, no H)
C) $-\text{OH}$
D) $-\text{COOH}$
E) $-\text{NH}_2$**
44. For the equilibrium $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H < 0$. Increasing temperature will:
**A) Shift the equilibrium to the right
B) Increase SO_3 yield only
C) Shift the equilibrium to the left (favours endothermic direction)
D) Have no effect on position
E) Stop the reaction**
45. The isotope ^{14}C has how many neutrons? (C atomic number = 6.)
A) 6 B) 7 C) 8 D) 14 E) 20
46. A catalyst increases the rate of a reaction by:
**A) Increasing the temperature
B) Providing an alternative pathway with lower activation energy
C) Being consumed in the net reaction
D) Increasing the activation energy
E) Shifting equilibrium toward products**
47. What type of reaction is: $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ (in light)?
A) Addition B) Combustion C) Neutralisation D) Elimination E) Substitution

Section D - Mathematics (Q48-54)

48. Solve: $(x - 2)(x + 3) = 0$.

- A) $x = 6$
- B) $x = -6$
- C) $x = 2$ only
- D) $x = -3$ only
- E) $x = 2$ or $x = -3$

49. Solve: $x^2 - 5x + 6 = 0$.

- A) $x = 2, 3$ B) $x = 1, 5$ C) $x = -2, -3$ D) $x = 1, 6$ E) No real roots

50. Solve: $3/x = 9$. ($x \neq 0$.)

- A) $x = 27$ B) $x = 1/3$ C) $x = 1/9$ D) $x = 3$ E) $x = -3$

51. Solve the inequality: $-2x + 1 > 7$.

- A) $x > 3$ B) $x > -3$ C) $x = -3$ D) $x < -3$ E) $x < 3$

52. In a right triangle with legs 6 and 8, what is the hypotenuse?

- A) 9 B) 10 C) 11 D) 12 E) 14

53. $\sin 30^\circ = ?$

- A) $\sqrt{2}/2$ B) 0 C) 1 D) $\sqrt{3}/2$ E) $1/2$

54. A is the set $\{1, 2, 3\}$; B is the set $\{a, b\}$. How many ordered pairs are in $A \times B$ (the Cartesian product)?

- A) 2 B) 3 C) 5 D) 6 E) 9

Section E - Physics (Q55-60)

55. A body accelerates uniformly from rest at 2 m/s^2 for 5 s. What distance does it cover?

- A) 10 m B) 20 m C) 25 m D) 50 m E) 100 m

56. Two resistors, 4Ω and 6Ω , are in series across a 10 V battery. What is the current?

- A) 0.5 A B) 1 A C) 2 A D) 2.5 A E) 10 A

57. A 1.5 V cell delivers 0.3 A through a lamp. What is the lamp's resistance?

- A) 0.2Ω B) 0.45Ω C) 2Ω D) 4.5Ω E) 5Ω

58. A 2 kg block is pushed with a force of 10 N on a frictionless surface. What is its acceleration?

- A) 2 m/s^2 B) 5 m/s^2 C) 8 m/s^2 D) 10 m/s^2 E) 20 m/s^2

59. A 5 kg object is lifted 2 m. How much gravitational PE is gained? ($g = 9.8 \text{ m/s}^2$.)

- A) 9.8 J B) 10 J C) 49 J D) 98 J E) 196 J

60. The acceleration due to gravity on the Moon is about $1/6$ of that on Earth. An astronaut's weight on the Moon compared to Earth is:

- A) $1/6$ as great
- B) $1/36$ as great
- C) 6 times greater
- D) The same
- E) Zero

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	A	2	B	3	C	4	E	5	D
6	C	7	C	8	E	9	C	10	D
11	A	12	D	13	B	14	A	15	E
16	B	17	A	18	B	19	D	20	E
21	D	22	C	23	B	24	A	25	E
26	B	27	D	28	C	29	E	30	A
31	B	32	D	33	C	34	E	35	D
36	A	37	A	38	B	39	D	40	D
41	E	42	C	43	A	44	C	45	C
46	B	47	E	48	E	49	A	50	B
51	D	52	B	53	E	54	D	55	C
56	B	57	E	58	B	59	D	60	A

Worked solutions

Each solution shows the reasoning that produces the key, and names the wrong options by their content. Letters refer to the option order printed in this paper.

Question 1 • Answer: A

The passage says each course of antibiotics applies selection favouring resistance and that resistance spreads faster where antibiotic use is high, so cutting unnecessary use removes that pressure and slows the spread. The claim that antibiotic use creates genetic mutations directly is unsupported: the changes arise by chance or gene transfer, and selection merely favours the carriers. The claims that resistance cannot arise without horizontal gene transfer, that all bacteria become resistant after one course, and that resistance is independent of selection pressure each either contradict the passage or overreach it with absolutes.

Question 2 • Answer: B

The passage defines negative feedback as a loop whose effector returns the variable to its set point, so the statement that negative feedback returns a variable toward its set point matches the text exactly. The claim that positive feedback is the main mechanism contradicts the passage, as does the claim that the effector pushes the variable further from the set point. The text explicitly lists body temperature among regulated variables and explicitly requires a sensor, so both remaining options are also false.

Question 3 • Answer: C

The decisive fact: Andreas Vesalius published *De humani corporis fabrica* in 1543, based on his own dissections, correcting many errors inherited from Galen. Leonardo dissected but published no such anatomical treatise; William Harvey, who described blood circulation, was a 17th century Englishman; Malpighi is associated with microscopy and capillaries, and Galvani with bioelectricity in frog muscle.

Question 4 • Answer: E

From P above Q and P below R we get the chain R above P above Q; adding S below Q extends it to R above P above Q above S. The ranking is fully determined, so the lowest score belongs to S, and the data are sufficient to rule out the undetermined answer.

Question 5 • Answer: D

This is modus tollens: from 'if it rains the match is cancelled' and 'the match was not cancelled', the only valid inference is that it did not rain. The claim that it rained would contradict the conditional, since rain would have forced cancellation, and 'it might have rained' is excluded for the same reason. The teams' refusal is not part of any premise.

Question 6 • Answer: C

The rule is $\text{output} = x^2 - 4$. Setting $x^2 - 4 = 32$ gives $x^2 = 36$, so $x = 6$ or $x = -6$; of the offered numbers only 6 works (check: $36 - 4 = 32$). Neighbours fail quickly: $5^2 - 4 = 21$ and $7^2 - 4 = 45$.

Question 7 • Answer: C

Step 1: the minute hand moves 6° per minute, so at 45 minutes it sits at $45 \times 6 = 270^\circ$ measured from 12. Step 2: the hour hand moves 30° per hour plus 0.5° per minute, so at 9:45 it is at $9 \times 30 + 45 \times 0.5 = 270 + 22.5 = 292.5^\circ$. Step 3: the angle between the hands is $292.5 - 270 = 22.5^\circ$. The tempting '0°' answer comes from seeing both hands near the 9, but the minute hand sits exactly on the 9 while the hour hand is three quarters of the way toward 10.

Question 8 • Answer: E

If A is B's father and B is C's sister, then B and C are siblings sharing the same father, so A is C's father as well. Uncle, grandfather, brother and cousin would each require an extra generation or a different relationship link that the statements do not provide.

Question 9 • Answer: C

The terms are perfect squares: $1 = 1^2$, $4 = 2^2$, $9 = 3^2$, $16 = 4^2$, $25 = 5^2$, so the next term is $6^2 = 36$. Cross-check with the differences, which grow by 2 each time (3, 5, 7, 9): the next difference is 11, and $25 + 11 = 36$. The value 49 is the square after next (7^2) and would skip a term.

Question 10 • Answer: D

Glycolysis takes place in the cytosol, where its soluble enzymes convert glucose to pyruvate without any membrane-bound organelle. The mitochondrial matrix hosts the Krebs cycle and the inner mitochondrial membrane the electron transport chain, so those locations belong to later stages of respiration.

Question 11 • Answer: A

When oxygen is scarce, human muscle runs lactic acid fermentation: lactate dehydrogenase reduces pyruvate to lactate, regenerating NAD^+ so glycolysis can continue producing ATP. Ethanol and CO_2 is the yeast (alcoholic) route, which humans lack, and converting pyruvate to acetyl-CoA is an aerobic mitochondrial step requiring oxygen-linked oxidative phosphorylation.

Question 12 • Answer: D

Modern estimates give about 30 to 32 ATP per glucose in eukaryotes: roughly 2 from glycolysis, about 2 directly from the Krebs cycle, and the remainder from oxidative phosphorylation, after accounting for proton leak and the cost of shuttling cytosolic NADH into the mitochondrion. The older textbook figure of 36 to 38 assumed perfect efficiency and is now considered too high, while 2 is only the glycolysis yield.

Question 13 • Answer: B

The coenzymes that ferry reducing power from glycolysis and the Krebs cycle to the respiratory chain are NAD^+/NADH and FAD/FADH_2 : NADH hands electrons to complex I and FADH_2 to complex II. Coenzyme A only escorts acetyl groups into the Krebs cycle, and ATP, ADP and AMP are energy-currency molecules, not electron carriers.

Question 14 • Answer: A

Electron transport complexes I, III and IV use the energy released by electron transfer to pump H^+ from the matrix into the intermembrane space; the resulting electrochemical gradient of protons is the proton-motive force that ATP synthase then exploits. ATP hydrolysis would run ATP synthase in reverse, and oxygen diffusion, pyruvate transport or Ca^{2+} release do not build this gradient.

Question 15 • Answer: E

Both respiration and photosynthesis run an electron transport chain coupled to chemiosmotic ATP synthesis, in the inner mitochondrial membrane and the thylakoid membrane respectively. The Calvin cycle and the light reactions belong only to photosynthesis, and the Krebs cycle only to respiration, so the feature shared by both processes is the chemiosmosis-driven electron transport chain.

Question 16 • Answer: B

Photosynthesis occurs in the chloroplast: the thylakoid membranes host the light reactions and the stroma the Calvin cycle. Mitochondria carry out respiration, the vacuole stores sap, and neither the nucleus nor peroxisomes conduct photosynthesis.

Question 17 • Answer: A

The light reactions split water, releasing O_2 , and use the captured light energy to synthesise ATP and NADPH; no sugar is produced at this stage. Glucose appears only later, when the Calvin cycle consumes that ATP and NADPH to fix CO_2 , so the claim that light reactions directly produce glucose and O_2 is the classic trap that blurs this division of labour.

Question 18 • Answer: B

Pulmonary veins carry freshly oxygenated blood from the lungs to the left atrium, one of the exceptions to the usual vein pattern. The pulmonary arteries do the opposite, taking deoxygenated blood from the right ventricle to the lungs, and the vena cavae return deoxygenated body blood to the right atrium; the aorta and coronary vessels are systemic, not lung-to-heart, vessels.

Question 19 • Answer: D

Gas exchange occurs in the alveoli: walls one cell thick and an enormous total surface area minimise the diffusion path for O₂ in and CO₂ out. The bronchi, trachea and pharynx are conducting airways with thick, cartilage- or mucosa-lined walls, and the aorta is a blood vessel, not an airway structure.

Question 20 • Answer: E

The parietal (oxyntic) cells of the gastric glands secrete hydrochloric acid and intrinsic factor. Chief cells secrete pepsinogen, goblet and mucous neck cells secrete mucus, and enterocytes are absorptive cells of the intestine, not acid secretors.

Question 21 • Answer: D

Bile is synthesised by hepatocytes in the liver and then concentrated and stored in the gallbladder, from which it is released into the duodenum to emulsify fats. The pancreas makes digestive enzymes and bicarbonate but not bile, and the reversed pairing that has the gallbladder producing bile and the liver storing it swaps the true roles.

Question 22 • Answer: C

The nephron is the kidney's functional unit: about one million per kidney, each performing filtration, reabsorption and secretion. The alveolus belongs to the lung, the sarcomere to muscle, the neuron to nervous tissue, and the hepatocyte to the liver.

Question 23 • Answer: B

Blood is filtered where the glomerulus sits inside Bowman's capsule: capillary blood pressure drives plasma across the filtration barrier into the capsular space. The loop of Henle and the collecting duct handle water and salt recovery downstream, the distal convoluted tubule fine-tunes ions, and the renal pelvis merely collects urine.

Question 24 • Answer: A

The lining of the stomach and intestines is simple columnar epithelium, typically with microvilli and goblet cells, suited to secretion and absorption. Stratified squamous epithelium is the wear-resistant lining of skin and oesophagus, pseudostratified ciliated epithelium lines the airways, transitional epithelium lines the bladder, and simple cuboidal epithelium occurs in ducts and tubules.

Question 25 • Answer: E

Tendons are cords of dense connective tissue, not muscle, and bone is attached by skeletal muscle tissue, so skeletal attachment is the exception. Smooth muscle does occur in the walls of blood vessels, intestines, uterus and bronchi, which is exactly its hollow-organ and vessel pattern.

Question 26 • Answer: B

Fibroblasts secrete the collagen (and elastin) fibres that dominate dense regular connective tissue such as tendons. Chondrocytes sit in cartilage, osteocytes in bone, adipocytes store fat, and erythrocytes transport oxygen; none of them builds tendon fibres.

Question 27 • Answer: D

Ventilation is driven mainly by rising arterial CO₂, which raises blood H⁺; central chemoreceptors in the medulla and peripheral chemoreceptors in the carotid and aortic bodies sense this and increase breathing. The claim that rising blood O₂ is the primary stimulus is doubly wrong: arterial O₂ hardly changes during moderate exercise, and O₂ becomes the decisive drive only when it falls substantially.

Question 28 • Answer: C

Insulin, released when blood glucose is high, promotes GLUT4-mediated glucose uptake in muscle and stimulates glycogen synthesis (glycogenesis) in liver and muscle. It suppresses the opposite processes: glycogenolysis, gluconeogenesis, lipolysis and ketogenesis belong to the low-insulin, glucagon-dominated state, and releasing glucose into the blood is what insulin prevents.

Question 29 • Answer: E

The thigh bone is the femur, the longest and strongest bone in the body. The tibia and fibula are lower-leg bones, the humerus is the arm bone, and the patella is the kneecap, a sesamoid bone.

Question 30 • Answer: A

A hypertonic solution has a higher solute concentration than the cell interior, so water leaves the red cell by osmosis and it shrinks, a change called crenation. Swelling and bursting is the fate in a hypotonic solution, where water flows inward, and an isotonic medium leaves the cell unchanged; the other outcomes (division, changing cell type) are impossible for a mature red cell.

Question 31 • Answer: B

Chromosomes align on the equatorial metaphase plate during metaphase, attached by spindle fibres from both poles. Anaphase is their separation toward the poles, prophase their condensation, telophase the reformation of nuclei, and cytokinesis the division of the cytoplasm.

Question 32 • Answer: D

In double-stranded DNA, adenine pairs with thymine (two hydrogen bonds) and guanine with cytosine (three hydrogen bonds). The pairing of adenine with uracil applies to RNA, where uracil replaces thymine, not to a DNA-DNA duplex, and pairing a base with itself or with the wrong-size partner would distort the helix's uniform width and bonding chemistry.

Question 33 • Answer: C

Balance the atoms of the product Al_2O_3 : taking 2 Al_2O_3 accounts for 4 Al and 6 O, so the reactants need 4 Al and 3 O_2 . The balanced equation is $4 \text{Al} + 3 \text{O}_2 \rightarrow 2 \text{Al}_2\text{O}_3$, hence the coefficient of O_2 is 3 (check: 6 oxygen atoms on each side).

Question 34 • Answer: E

Oxidation numbers sum to zero in a neutral salt: K contributes +1 and each of the four oxygens -2 , so $+1 + x + 4 \times (-2) = 0$, giving $x = +7$ for manganese. The value +4 would balance only two oxygens (as in MnO_2), and -1 confuses oxygen's charge with manganese's oxidation state.

Question 35 • Answer: D

Step 1: moles of $\text{HCl} = 0.0250 \text{ L} \times 0.10 \text{ mol/L} = 0.0025 \text{ mol}$. Step 2: $\text{HCl} + \text{NaOH}$ reacts 1:1, so 0.0025 mol of NaOH is required. Step 3: volume of $\text{NaOH} = 0.0025 \text{ mol} \div 0.05 \text{ mol/L} = 0.050 \text{ L} = 50 \text{ mL}$. The 25 mL answer forgets that the NaOH is half as concentrated as the acid, so twice the acid volume is needed.

Question 36 • Answer: A

A buffer pairs a weak acid with its conjugate base (or a weak base with its conjugate acid), such as acetic acid and acetate: the weak-acid partner neutralises added base and the conjugate base neutralises added acid. A strong acid is fully dissociated and has no conjugate reserve to draw on, and mixing a strong acid with a strong base simply neutralises rather than buffers.

Question 37 • Answer: A

Electronegativity increases toward the top right of the periodic table, and fluorine tops the entire scale at 4.0 on the Pauling scale. Oxygen is the highest among the rest, chlorine follows, while sodium and caesium sit at the electropositive bottom-left extreme.

Question 38 • Answer: B

Carbon in CH_4 has four bonding pairs and no lone pairs; VSEPR pushes them as far apart as possible, giving a tetrahedron with bond angles of 109.5° . Pyramidal geometry would require a lone pair (as in NH_3), and linear or trigonal planar shapes correspond to two or three electron domains, not four.

Question 39 • Answer: D

The standard molar volume of an ideal gas at STP (0°C , 1 atm) is 22.4 L per mole: $V = nRT/P = 1 \times 0.0821 \times 273 \div 1 \approx 22.4 \text{ L}$. The 24.0 L figure corresponds to room temperature (about 25°C), not STP, and 11.2 L is the volume of half a mole.

Question 40 • Answer: D

Molar mass of $\text{CaCO}_3 = 40 + 12 + (3 \times 16) = 40 + 12 + 48 = 100 \text{ g/mol}$. The values 68 and 56 arise from dropping or short-changing the three oxygens, and 84 and 116 match no correct combination of the given atomic masses.

Question 41 • Answer: E

pH measures H^+ concentration: 0.1 M HCl, being a strong acid, fully dissociates to give $[H^+] = 0.1$ M, so $pH = 1$, the lowest on offer. 0.01 M HCl gives pH 2, and 0.1 M acetic acid dissociates only partially, landing near pH 2.9; pure water is neutral at pH 7 and 0.1 M NaOH is strongly alkaline (about pH 13).

Question 42 • Answer: C

Aromaticity requires a cyclic, conjugated π system satisfying Hückel's rule, and benzene (C_6H_6) with its six delocalised π electrons is the archetype. Ethene, ethanol, methane and acetylene are all acyclic molecules without a ring or aromatic sextet, so none can be aromatic.

Question 43 • Answer: A

Aldehydes carry the formyl group, $-CHO$: a carbonyl carbon bonded to at least one hydrogen atom, as in ethanal CH_3CHO . The $-OH$ group marks alcohols, $-COOH$ carboxylic acids, $-NH_2$ amines, and an internal $-C=O$ carrying no hydrogen defines a ketone instead.

Question 44 • Answer: C

For $2SO_2 + O_2 \rightleftharpoons 2SO_3$ with $\Delta H < 0$, the forward reaction is exothermic, so the reverse is endothermic. By Le Chatelier's principle, raising the temperature favours the direction that absorbs the added heat, that is the reverse, shifting the equilibrium left and reducing SO_3 yield. The tempting 'shift to the right' answer would hold only if the forward reaction were endothermic.

Question 45 • Answer: C

Neutrons = mass number – atomic number = $14 - 6 = 8$, since carbon always has 6 protons. The answer 14 mistakes the mass number for the neutron count, and 6 counts protons instead of neutrons.

Question 46 • Answer: B

A catalyst provides an alternative reaction pathway with a lower activation energy, so a larger fraction of collisions succeeds and the rate rises; the catalyst is regenerated, not consumed. It does not heat the reaction, it lowers rather than raises the barrier, and because it speeds forward and reverse directions equally it never shifts the equilibrium toward products.

Question 47 • Answer: E

In light, a chlorine radical replaces a hydrogen atom of methane, giving CH_3Cl plus HCl; one atom swaps partners, which is free-radical substitution. Addition requires a double or triple bond to open, which methane lacks, and nothing is eliminated, neutralised or burned here.

Question 48 • Answer: E

A product is zero exactly when at least one factor is zero: $x - 2 = 0$ gives $x = 2$, and $x + 3 = 0$ gives $x = -3$, so both roots must be stated. Keeping only one of them discards a valid solution, and 6 or -6 satisfy no factor of this equation.

Question 49 • Answer: A

Factorise: $x^2 - 5x + 6 = (x - 2)(x - 3)$, so $x = 2$ or $x = 3$ (check: $2 + 3 = 5$ and $2 \times 3 = 6$). The pair -2 and -3 would produce a middle term of $+5x$, and 1 and 6 multiply to 6 but sum to 7, not 5.

Question 50 • Answer: B

Multiply both sides of $3/x = 9$ by x : $3 = 9x$, hence $x = 3/9 = 1/3$. The value 27 comes from multiplying instead of dividing, and 3 from mishandling the fraction; substitution confirms $3 \div (1/3) = 9$.

Question 51 • Answer: D

Step 1: subtract 1 from both sides to get $-2x > 6$. Step 2: divide by -2 , which flips the inequality sign, giving $x < -3$ (check $x = -4$: $-2(-4) + 1 = 9 > 7$ holds, while $x = 0$ gives 1, which does not). Forgetting the sign flip produces the mirror-image wrong answers $x > 3$ or $x > -3$.

Question 52 • Answer: B

Pythagoras' theorem gives hypotenuse = $\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$, the classic 6-8-10 triple (a scaled 3-4-5 triangle). Any value below 10 cannot be the hypotenuse, since the hypotenuse must be the longest side of a right triangle.

Question 53 • Answer: E

On the unit circle, or from a 30-60-90 triangle with sides 1, $\sqrt{3}$ and 2, $\sin 30^\circ = \text{opposite/hypotenuse} = 1/2$. The value $\sqrt{3}/2$ belongs to $\sin 60^\circ$ (and to $\cos 30^\circ$), while 0 and 1 are the sines of 0° and 90° .

Question 54 • Answer: D

Each of the 3 elements of A pairs with each of the 2 elements of B, so $|A \times B| = 3 \times 2 = 6$ ordered pairs: (1,a), (1,b), (2,a), (2,b), (3,a), (3,b). Adding the set sizes ($3 + 2 = 5$) confuses the size of a union with that of a Cartesian product, and 9 would be the size of $A \times A$, not $A \times B$.

Question 55 • Answer: C

For uniform acceleration from rest, $s = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times 5^2 = \frac{1}{2} \times 2 \times 25 = 25$ m. The 10 m answer is actually the speed reached after 5 s ($v = at = 10$ m/s) mistaken for a distance, and 50 m omits the factor $\frac{1}{2}$.

Question 56 • Answer: B

Series resistances add: $R = 4 + 6 = 10 \Omega$. Ohm's law then gives $I = V/R = 10 \text{ V} \div 10 \Omega = 1 \text{ A}$. The other answers come from dividing by a single resistor ($10 \text{ V} \div 4 \Omega$ or $10 \text{ V} \div 6 \Omega$) or from wrongly treating the resistors as parallel.

Question 57 • Answer: E

Ohm's law: $R = V/I = 1.5 \text{ V} \div 0.3 \text{ A} = 5 \Omega$. The value 0.45Ω comes from multiplying V by I (0.45 is actually the power in watts), and 0.2 from dividing I by V instead of V by I.

Question 58 • Answer: B

Newton's second law: $a = F/m = 10 \text{ N} \div 2 \text{ kg} = 5 \text{ m/s}^2$. Multiplying (giving 20 m/s^2) or dividing the other way (giving 0.2 m/s^2) misapplies the relation; since the surface is frictionless, the full 10 N accelerates the block.

Question 59 • Answer: D

Gravitational potential energy gained = $mgh = 5 \text{ kg} \times 9.8 \text{ m/s}^2 \times 2 \text{ m} = 98 \text{ J}$. The 9.8 J value is just g in numeric form, and 49 J uses 5×9.8 but drops the 2 m height.

Question 60 • Answer: A

Weight is $W = mg$, so with the Moon's gravitational acceleration one sixth of Earth's, the astronaut's weight is likewise one sixth of its Earth value; mass stays the same, but weight scales with g. The $1/36$ figure wrongly squares the factor, and weight on the Moon is reduced, not zero.